

open-loop model. The resulting open-loop model is seen in Fig. 31.16. We will initially assume that r_o for the MOSFETS is much larger than the discrete resistors and that the bulk and source are tied together ($v_{sb} = 0$). As we progress through the chapter, more difficult circuits will include drain-to-source resistances in our small-signal analysis.

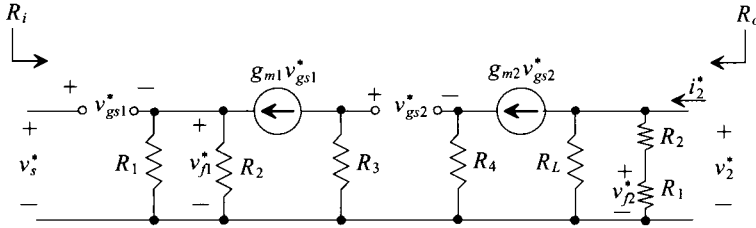


Figure 31.16 Open-loop small-signal model of Fig. 31.13.

The open-loop model is now ready to be analyzed in order to calculate A_{OL} . Since we are using a series (voltage)-shunt (voltage) feedback amplifier, the units of A_{OL} will be V/V and

$$A_{OL} = \frac{v_2^*}{v_s^*} \quad (31.34)$$

Solving for A_{OL} yields

$$A_{OL} = \frac{v_2^*}{v_s^*} = \left(\frac{v_2^*}{v_{gs2}^*} \right) \left(\frac{v_{gs2}^*}{v_{gs1}^*} \right) \left(\frac{v_{gs1}^*}{v_s^*} \right) = [-g_{m2}R_L || (R_2 + R_1)] \left[-\frac{g_{m1}R_3}{1 + g_{m2}R_4} \right] \left[\frac{1}{1 + g_{m1}(R_1 || R_2)} \right] \quad (31.35)$$

Next, the value of β can also be calculated from the open-loop model. Remembering that β is defined as the gain from the output back to the input mixing variable, v_f , we can write

$$\beta = \frac{v_f^*}{v_2^*} = \frac{R_1}{R_1 + R_2} \quad (31.36)$$

since the β network is simply a voltage divider relationship. Notice that the open-loop circuit now contains two values of R_2 and v_{f1}^* . In this example, since r_o was assumed to be infinite, the gain from v_2^* to v_{f1}^* will be zero. If r_o had not been neglected, the gain from v_2^* to v_{f1}^* would have been small but finite. Therefore, it can be said that a reverse path exists through the basic amplifier as well as through the feedback network. However, the gain from v_2^* to v_{f2}^* , though less than one, will be significantly larger than from v_2^* to v_{f1}^* . *Therefore, just as the forward path through the feedback network was neglected, the reverse path through the basic amplifier is assumed to be much smaller than the reverse path through the feedback path. Therefore, the value of β is calculated using the resistor, R_2 , closest to the output.*

Next, the value for R_i and R_o will be calculated. These values are determined using the open-loop model generated in Fig. 31.16. Since we are using MOS devices, it should

be obvious that the input resistance to the open-loop circuit is ∞ . The output resistance, however, can be calculated shorting the gate of M1 to ground and applying a test voltage to the output. Since the input is grounded, $v_{g2} = 0$, and R_o is simply the parallel combination of resistances at the output (assuming that r_{o2} is very large).

$$R_o = \frac{v_o^*}{i_2^*} = R_L || (R_1 + R_2) \quad (31.37)$$

Once the open-loop values are calculated, the closed-loop values are easily attained. The closed-loop values are

$$A_{CL} = \frac{v_2}{v_s} = \frac{A_{OL}}{1 + A_{OL}\beta} \quad (31.38)$$

$$R_{inf} = \frac{v_s}{i_s} = R_i(1 + A_{OL}\beta) \quad (31.39)$$

$$R_{out} = R_{of} = \frac{v_2}{i_2} = \frac{R_o}{1 + A_{OL}\beta} \quad (31.40)$$

Analysis of this problem has not yet been completed. Notice that we initially neglected the source resistance and the biasing resistors, R_{G1} and R_{G2} , since they played no part in the feedback analysis of the amplifier. However, they will have an effect in the overall gain and need to be considered. The total gain of the entire circuit is

$$\frac{v_2}{v_1} = \frac{v_s}{v_1} \cdot \frac{v_2}{v_s} = \frac{R_{G1} || R_{G2}}{R_{G1} || R_{G2} + R_s} \cdot A_{CL} \quad (31.41)$$

and the value of R_{in} as seen by the signal source is

$$R_{in} = \frac{v_1}{i_1} = R_{G1} || R_{G2} \quad (31.42)$$

and the analysis is now complete. Notice that the value of R_{in} is not the same as R_{inf} .

Example 31.1

For the series-shunt circuit shown in Fig. 31.17a, draw the closed-loop small-signal model and identify the forward and feedback paths, draw the open-loop model, calculate A_{OL} , β , R_i , and R_o , and calculate the closed-loop parameters v_2/v_1 , v_1/i_1 , and v_2/i_2 . You may assume that $r_o = \infty$ for both devices and that DC analysis has already been performed with $g_{m1} = g_{m2} = 1 \text{ mA/V}$.

The closed-loop small-signal model can be seen in Fig. 31.17b with the forward and feedback paths drawn. The open-loop model with loading effects is seen in Fig. 31.17c. Since the output uses shunt sampling, the value of $R_{\beta i}$ is found by shorting the output node and looking at the load resistance resulting from the feedback network. In this case, $R_{\beta i}$ is equal to R_2 . Since the input mixing is series, the input device, M1, is "severed," so that only $R_1 + R_2$ is seen looking into the feedback loop from the output. Thus, $R_{\beta o} = R_1 + R_2$.

The values of r_o are considered to be infinite, so the values of R_i and R_o can be found by inspection, $R_i = \infty$, and $R_o = R_1 + R_2 = 11 \text{ k}\Omega$. Notice that when solving the open-loop circuit (Fig. 31.17c) v_1 and R_G are not included since we are only interested in solving the feedback portion of the circuit.

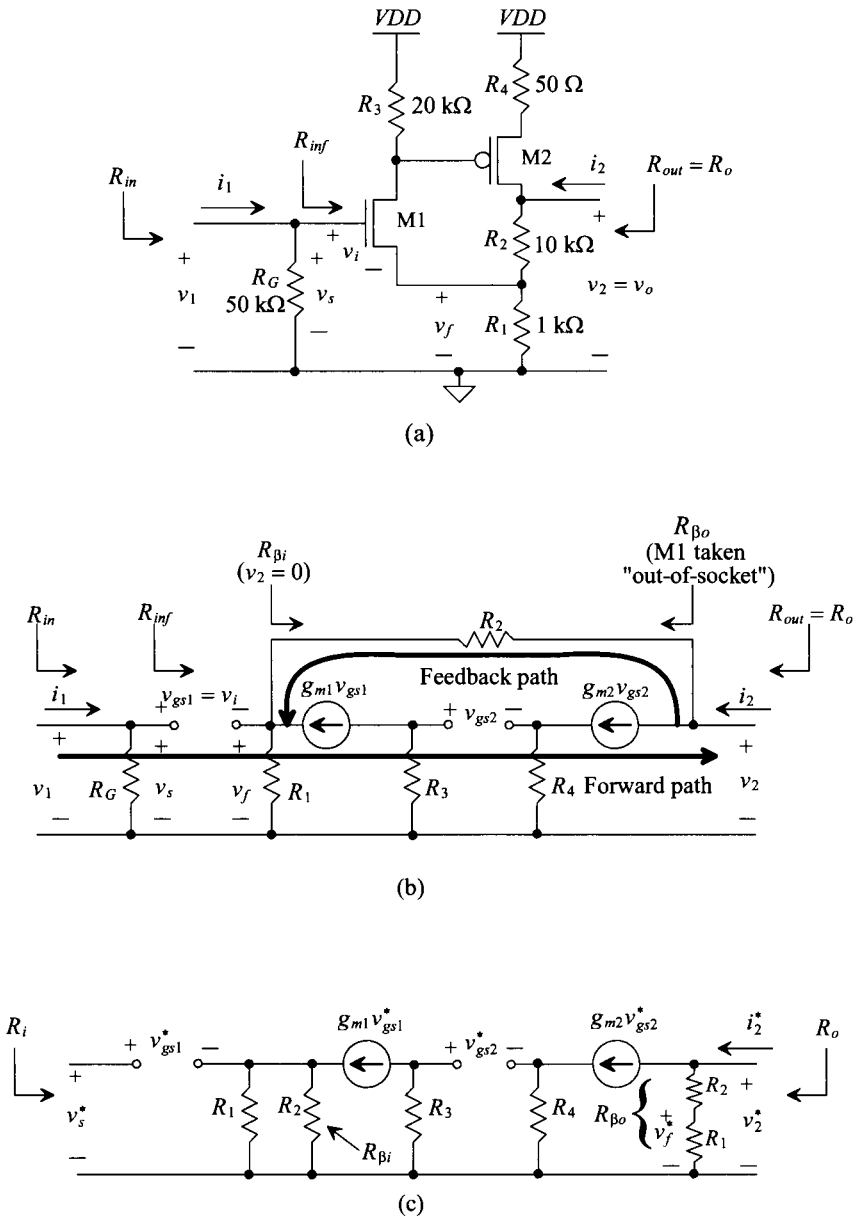


Figure 31.17 (a) Series-shunt circuit used in Ex. 31.1; (b) its closed-loop small-signal model; and (c) the resulting open-loop model.

The open-loop gain, A_{OL} , can be found simply as the gain of two common source amplifiers with source resistance,

$$A_{OL} = \frac{v_2^*}{v_s^*} = \frac{v_2^*}{v_{g2}^*} \cdot \frac{v_{g2}^*}{v_s^*} = \left[\frac{-g_{m2}(R_2 + R_1)}{1 + g_{m2}R_4} \right] \left[\frac{-g_{m1}R_3}{1 + g_{m1}(R_1 || R_2)} \right] = 109.8 \text{ V/V}$$

The value of β is always the gain from the output to the feedback variable. In this case, a simple voltage divider relationship exists such that

$$\beta = \frac{v_f^*}{v_2^*} = \frac{R_1}{R_1 + R_2} = 0.0909 \text{ V/V}$$

Again, notice that the product of $A_{OL}\beta$ is positive and unitless. Now that the open-loop values are calculated, the closed-loop values can be found by using Eqs. (31.38) - (31.40).

$$A_{CL} = \frac{v_2}{v_s} = \frac{A_{OL}}{1 + A_{OL}\beta} = \frac{109.8}{1 + (109.8 \cdot 0.0909)} = 10 \text{ V/V}$$

A simple check will verify that the solution is correct if $A_{CL} \approx \frac{1}{\beta}$, which is the case.

Notice in Fig. 31.17a and Fig. 31.17b that closed-loop output impedance, R_{of} is equal to the value of R_{out} . However, the value of the closed-loop input impedance, R_{inf} , is not equal to R_{in} , since the feedback amplifier itself excludes the input source and the gate resistor, R_G . The closed-loop values, R_{inf} , and R_{of} , are also easily attained, that is,

$$R_{inf} = \infty \text{ (since } R_i = \infty \text{)}$$

and

$$R_{of} = R_{out} = \frac{v_2}{i_2} = \frac{R_o}{(1 + A_{OL}\beta)} = \frac{11 \text{ k}\Omega}{10.98} = 1.002 \Omega$$

The last step involves finding R_{in} and v_2/v_1 . The value of R_{in} can be found by examining Fig. 31.17a as,

$$R_{in} = \frac{v_1}{i_1} = R_G || R_{inf} = 50 \text{ k}\Omega$$

The overall gain, v_2/v_1 , will be equal to the value of A_{CL} since the input voltage, v_1 is equal to v_s . If we had included a value for a source resistance, then an additional voltage divider relationship would have been needed to formulate v_2/v_1 as seen in Eq. (31.41)

$$\frac{v_2}{v_1} = A_{CL} = 10 \text{ V/V} \blacksquare$$

31.5 The Transimpedance Amp (Shunt-Shunt Feedback)

Shunt-shunt feedback mixes current at its input and samples voltage at the output. Consider the ideal shunt-shunt feedback amplifier shown with open-loop values, A_{OL} , β , R_i , and R_o , in Fig. 31.18. In the ideal case, $R_{\beta i}$ is infinite and $R_{\beta o}$ is zero. Notice that the basic amplifier and the β network are in parallel at both the input and the output. The β

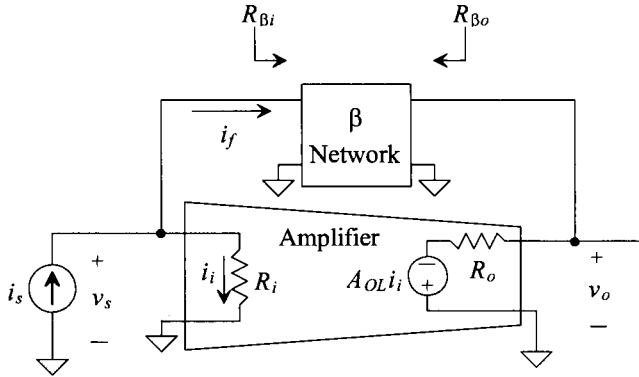


Figure 31.18 An ideal transimpedance (shunt-shunt) amplifier.

network shunts the basic amplifier; therefore, the input variable is a current. Looking into the output of the amplifier, we see that the feedback path is in parallel with or shunts the output signal. Therefore, the output signal is a voltage. Also note that since the input variable is a current and the output variable is a voltage, the basic amplifier has units of V/I, also known as a transimpedance amplifier. Since $A_{OL}\beta$ should always be unitless and positive, β will have units of I/V (mhos).

The closed-loop gain is

$$A_{CL} = \frac{v_o}{i_s} = \frac{A_{OL}}{1 + A_{OL}\beta} \quad (31.43)$$

Notice that the basic amplifier circuit is inverting, corresponding to an ideal op-amp circuit. Since A_{OL} is negative, then β must also be negative to ensure that negative feedback exists.

To calculate the input impedance of the transimpedance amplifier we apply a test current source to the input of the amplifier, such that $i_s = i_{test}$ and $v_s = v_{test}$. The test current and the input impedance are determined by

$$i_{test} = \frac{v_{test}}{R_i} + i_f = \frac{v_{test}}{R_i} + \beta v_o = \frac{v_{test}}{R_i} + \beta A_{OL} i_i = \frac{(1 + A_{OL}\beta) \cdot v_{test}}{R_i} \quad (31.44)$$

Assuming that the β network does not load the basic amplifier, we note that the closed-loop input impedance becomes

$$R_{inf} = \frac{v_{test}}{i_{test}} = \frac{R_i}{1 + A_{OL}\beta} \quad (31.45)$$

A similar analysis of the output impedance of the transimpedance amplifier shows that

$$R_{of} = \frac{R_o}{(1 + A_{OL}\beta)} \quad (31.46)$$

The ideal transimpedance amplifier has zero input resistance and zero output resistance. We can see from Eqs. (31.45) and (31.46) that the addition of the feedback helps to make the basic amplifier appear closer to the ideal.

Now examine the transistor level shunt-shunt feedback circuit shown in Fig. 31.19. The closed-loop and open-loop small-signal models are seen in Fig. 31.20 and Fig. 31.21, respectively. The gate is attached to AC ground since it is attached to a DC source and the DC current source I_{SS} is an AC open circuit. Interestingly, since the values of r_{o1} and r_{o2} have been included, there is a feedback path through the basic amplifier. However, this gain is very small compared to the feedback path through the β network and is typically assumed to be negligible.

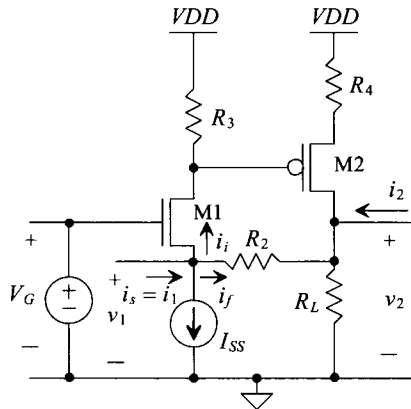


Figure 31.19 Shunt-shunt feedback amplifier.

The open-loop analysis begins by evaluating the effects of the β network on the basic amplifier using the rules stated earlier. The resistance, $R_{\beta i}$, will be the equivalent resistance seen looking into the β network from the input with the output node shorted to ground. The equivalent resistance looking into the β network from the output, $R_{\beta o}$, will be calculated with the input shorted to ground.

Since we are using shunt-shunt feedback, A_{OL} will be defined as

$$A_{OL} = \frac{v_2^*}{i_s^*} = \frac{v_2^*}{v_{g2}^*} \cdot \frac{v_{g2}^*}{v_1^*} \cdot \frac{v_1^*}{i_s^*} \quad (31.47)$$

Solving the first term in Eq. (31.47), can be quite extensive if using standard circuit analysis. A circuit technique based on two-port theory will greatly simplify the analysis. For example, the equivalent circuit for the gain, $\frac{v_2^*}{v_{g2}^*}$, can be seen in Fig. 31.22a. However, the circuit seen in Fig. 31.22b shows the equivalent circuit using an equivalent transconductance, G_M , and output resistance, R_{Leq} . The gain of the equivalent circuit, and hence the actual circuit, is

$$\frac{v_2^*}{v_{g2}^*} = G_M R_{Leq} \quad (31.48)$$

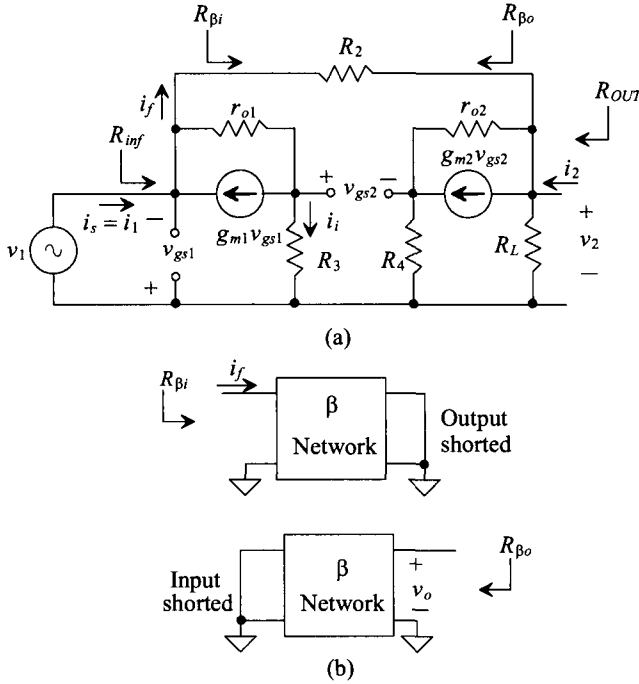


Figure 31.20 (a) Closed-loop small-signal model of Fig. 31.19 and (b) method for determining the feedback network loading.

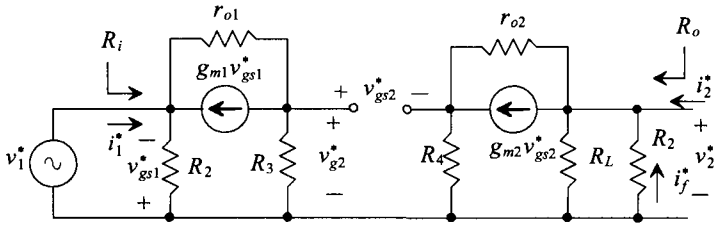


Figure 31.21 Open-loop small-signal model of Fig. 31.19.

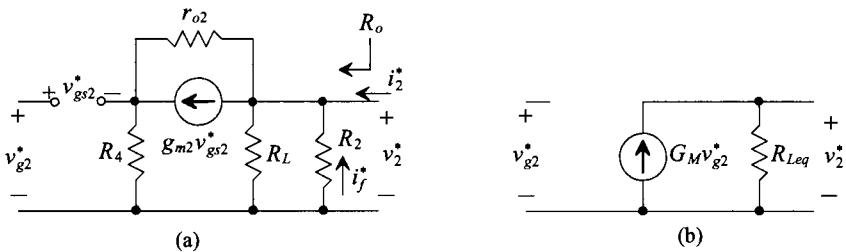


Figure 31.22 (a) Solving a portion of Fig. 31.21, including the drain-to-source resistance, and (b) the equivalent transconductance model.

The value of R_{Leq} can easily be found as

$$R_{Leq} = R_L || R_2 || R_{inD2} \quad (31.49)$$

where R_{inD2} is the resistance seen looking into the drain of M2. From Ch. 20, we know that this resistance is

$$R_{inD2} = [(1 + g_{m2}R_4)r_{o2} + R_4] \quad (31.50)$$

The value of G_M is the short-circuit transconductance and is defined as

$$G_M = \frac{i_o^*}{v_{g2}^*} (R_{Leq} = 0) \quad (31.51)$$

which means that the effective transconductance can be found by shorting the equivalent load resistance, in this case $R_L || R_2$, and finding the gain from the short-circuit current to the input voltage. As seen in Fig. 31.23, the equations used to find G_M are

$$i_o^* = -g_{m2}v_{gs2} + \frac{v_{s2}}{r_{o2}} \quad (31.52)$$

$$v_{s2}^* = -i_o R_4 \quad (31.53)$$

$$v_{s2}^* + v_{gs2}^* = v_{g2}^* \quad (31.54)$$

and solving Eqs. (31.52) - (31.54) yields

$$G_M = \frac{i_o^*}{v_{g2}^*} = \frac{-g_{m2}}{1 + g_{m2}R_4 + \frac{R_4}{r_{o2}}} \quad (31.55)$$

the gain, $\frac{v_2^*}{v_{g2}^*}$, becomes

$$\frac{v_2^*}{v_{g2}^*} = \frac{-g_{m2}(R_L || R_2 || [(1 + g_{m2}R_4)r_{o2} + R_4])}{1 + g_{m2}R_4 + \frac{R_4}{r_{o2}}} \quad (31.56)$$

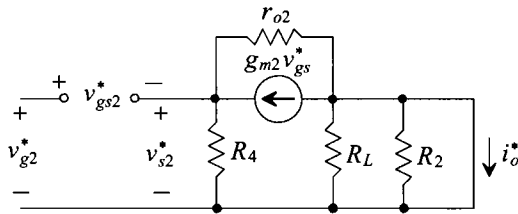


Figure 31.23 Circuit used to determine the equivalent transconductance.

Referring back to Eq. (31.47), the second factor, $\frac{v_{g2}^*}{v_1^*}$, can be found by analyzing Fig. 31.21 as

$$\frac{v_{g2}^*}{v_1^*} = \frac{g_{m1}R_3 + \frac{R_3}{r_{o1}}}{1 + \frac{R_3}{r_{o1}}} \quad (31.57)$$

The last term in Eq. (31.47) is simply the input resistance, R_i , of the open-loop circuit shown. Using a test source, we can determine this value to be

$$R_i = \frac{v_1^*}{i_s^*} = \frac{v_i}{i_i} || R_2 = \left(\frac{1 + \frac{R_3}{r_{o1}}}{g_{m1} + \frac{1}{r_{o1}}} \right) || R_2 \quad (31.58)$$

Therefore, the entire expression for the open-loop gain, A_{OL} , becomes

$$A_{OL} = \frac{v_2^*}{i_s^*} = \frac{-g_{m2}(R_L || R_2 || ((1 + g_{m2}R_4)r_{o2} + R_4))}{1 + g_{m2}R_4 + \frac{R_4}{r_{o2}}} \cdot \frac{g_{m1}R_3 + \frac{R_3}{r_{o1}}}{1 + \frac{R_3}{r_{o1}}} \cdot \left(\frac{1 + \frac{R_3}{r_{o1}}}{g_{m1} + \frac{1}{r_{o1}}} \right) || R_2 \Omega \quad (31.59)$$

At first glance, this equation may appear quite daunting. However, notice that if r_{o2} is much greater than R_2 , R_4 , and R_L and if r_{o1} is much greater than R_3 , the open-loop gain simplifies to

$$A_{OL} = [\text{common source amp with source resistance}][\text{common gate amp}][R_i] \\ \approx \frac{-g_{m2}(R_L || R_2)}{1 + g_{m2}R_4} \cdot g_{m1}R_3 \cdot \frac{1}{g_{m1}} || R_2 \Omega \quad (31.60)$$

Equation (31.60) is typically used for discrete designs in which higher values of currents are used, therefore requiring lower resistor values. However, active loads can easily be used, as seen in Fig. 31.24. Here, the resistor, r_{o3} now replaces R_3 . Since R_4 is a source degeneration resistor, its value will be small. The active load that replaces R_4 is a gate-drain connected device and equal to $\frac{1}{g_{m4}} || r_{o4}$. And the resistor, R_L , is now replaced by r_{o5} . Therefore, the open-loop gain of Fig. 31.24 can be written by using Eq. (31.29) and making the proper substitutions:

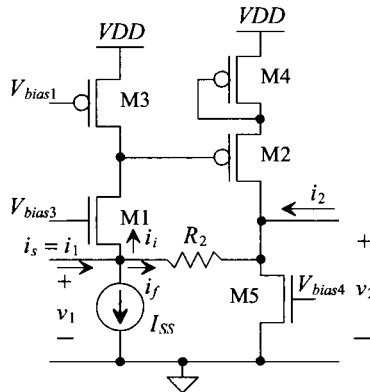


Figure 31.24 Shunt-shunt feedback amplifier using active loads.

$$A_{OL} = \frac{-g_{m2}(r_{o5} || R_2 || (1 + g_{m2}(\frac{1}{g_{m4}} || r_{o4}))r_{o2} + \frac{1}{g_{m4}} || r_{o4}))}{1 + g_{m2}(\frac{1}{g_{m4}} || r_{o4}) + \frac{\frac{1}{g_{m4}} || r_{o4}}{r_{o2}}} \cdot \frac{g_{m1}r_{o3} + \frac{r_{o3}}{r_{o1}}}{1 + \frac{r_{o3}}{r_{o1}}} \left[\frac{1 + \frac{r_{o3}}{r_{o1}}}{g_{m1} + \frac{1}{r_{o1}}} || R_2 \right] \quad (31.61)$$

which can be approximated as

$$A_{OL} = \frac{v_2^*}{i_s^*} \approx \frac{-g_{m2}(R_2)}{1 + g_{m2}(\frac{1}{g_{m4}})} \cdot \frac{1 + g_{m1}r_{o3}}{2} \cdot \left(\frac{1}{g_{m1}} || R_2 \right) \Omega \quad (31.62)$$

if it is assumed that $r_{o1} \approx r_{o3}$ and that the discrete resistor, R_2 , is relatively small compared to the impedances of the active loads. An active device could have been used to substitute even the resistor, R_2 . This will be discussed further later in the chapter.

Next, the value of β is calculated as

$$\beta = \frac{i_f^*}{v_2^*} = -\frac{1}{R_2} \text{ mhos} \quad (31.63)$$

Again, note that $A_{OL}\beta$ is unitless and overall positive.

Now that all of the open-loop parameters, A_{OL} , β , R_i , and R_o , have been found, the closed-loop values are easily calculated. The value for the closed-loop gain is

$$A_{CL} = \frac{v_2}{i_s} = \frac{A_{OL}}{1 + A_{OL}\beta} \quad (31.64)$$

The closed-loop input impedance is

$$R_{inf} = \frac{R_i}{1 + A_{OL}\beta} \quad (31.65)$$

and the value for the closed-loop output impedance is

$$R_{of} = R_{out} = \frac{R_o}{1 + A_{OL}\beta} \quad (31.66)$$

We should be able to see a trend in the feedback effects. Shunt mixing and shunt sampling cause the input and output impedances to decrease by the factor $(1 + A_{OL}\beta)$. Similarly, series input mixing and series sampling cause the input and output impedances to increase by the factor $(1 + A_{OL}\beta)$.

In many cases, the overall gain is expressed in terms of a voltage gain, $\frac{v_2}{v_1}$. Since we have calculated the transfer function in terms of the current, i_s , we can easily express this in terms of v_1 by

$$\frac{v_2}{v_1} = \frac{v_2}{i_s(R_{inf})} = A_{CL} \left(\frac{1}{R_{inf}} \right) \quad (31.67)$$

31.5.1 Simple Feedback Using a Gate-Drain Resistor

One of the most popular, and simplest, examples of shunt-shunt feedback is seen in Fig. 31.25 and consists of a simple inverting amplifier with a resistor connecting the gate and the drain. The feedback resistor, R_2 , is usually a large value and serves several important

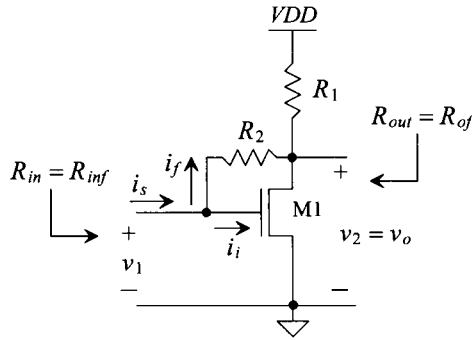


Figure 31.25 Shunt-shunt feedback using a simple gate-drain connected device.

functions. When analyzing the DC characteristics of this circuit, the voltage at the drain will be equal to the voltage on the gate, since there is no DC current flowing through R_2 (when the input is AC coupled). This ensures that the device is always in saturation and provides biasing with no other components needed. We will examine the effects of R_2 on the AC response of the amplifier and soon discover that it has little effect on the gain (if it is large) of the amplifier as well.

Figure 31.26a and b shows the small-signal models for the closed and open-loop circuits, respectively. Since the feedback resistor sums current at the gate of M1, the mixing circuit is shunt. And since the feedback is taken off of the same node as the output, the sampling circuit is shunt. Therefore, the value of $R_{\beta i}$ for the open-loop model is found by shorting the output and looking into the feedback loop from the input, and the value of $R_{\beta o}$ is found by shorting the input and looking into the feedback loop from the output.

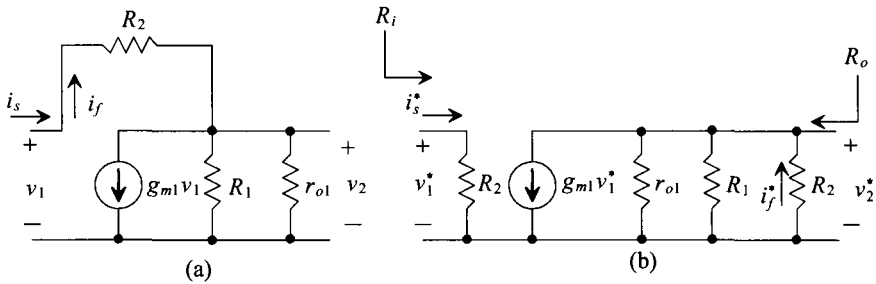


Figure 31.26 Small-signal model of Fig. 31.25: (a) the closed-loop model and (b) the open-loop model.

The open-loop values can be calculated as

$$A_{OL} = \frac{v_2^*}{i_s^*} = \frac{v_2^*}{v_1^*} \cdot \frac{v_1^*}{i_s^*} = [-g_{m1}(R_1 || R_2 || r_{o1})][R_2] \text{ V/A} \quad (31.68)$$

$$R_i = R_2 \quad (31.69)$$

$$R_o = R_1 || R_2 || r_{o1} \quad (31.70)$$